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SYSTEMATIC REVIEW

The Association Between the Acute:Chronic Workload Ratio and Injury and its Application in Team Sports: A Systematic Review

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Abstract

Background There has been a recent increase in research examining training load as a method of mitigating injury risk due to its known detrimental effects on player welfare and team performance. The acute:chronic workload ratio (ACWR) takes into account the current training load (acute) and the training load that an athlete has been prepared for (chronic). The ACWR can be calculated using; (1) the rolling average model (RA) and (2) the exponentially weighted moving average model (EWMA).

Objective The primary aim of this systematic review was to investigate the literature examining the association between the occurrence of injury and the ACWR and to investigate if sufficient evidence exists to determine the best method of application of the ACWR in team sports.

Methods Studies were identified through a comprehensive search of the following databases: EMBASE, Medline, SPORTDiscus, SCOPUS, AMED and CINAHL. Extensive data extraction was performed. The methodological quality of the included studies was assessed according to the Newcastle–Ottawa Scale (NOS) for Cohort Studies.

Results A total of 22 articles met the inclusion criteria. The assessment of article quality had an overall median NOS score of 8 (range 5–9). The findings of this review support the association between the ACWR and non-contact injuries and its use as a valuable tool for monitoring training load as part of a larger scale multifaceted monitoring system that includes other proven methods. There is support for both models, but the EWMA is the more suitable measure, in part due to its greater sensitivity. The most appropriate acute and chronic time periods, and training load variables, may be dependent on the specific sport and its structure.

Conclusions For practitioners, it is the important to understand the intricacies of the ACWR before deciding the best method of calculation. Future research needs to focus on the more sensitive EWMA model, for both sexes, across a larger range of sports and time frames and also combinations with other injury risk factors.

1 Introduction

Injury has a detrimental effect on player welfare, health and lifestyle including an increased risk of distress, anxiety, sleep disturbance, depression [1, 2], cognitive impairment [3, 4] and financial cost due to loss of employment [5]. Additionally, a recent systematic review has suggested that injuries have a negative impact on team sport performance, with increased player availability improving chances of success [6]. These associated detrimental effects, therefore, justify the recent increase in research examining associations with injury [7, 8].

Injury is multifactorial in nature with inherent risk factors that are both non-modifiable (e.g.: age, sex) and modifiable (e.g.: strength, flexibility) [9, 10]. The manipulation of modifiable risk factors provides coaches and sport scientists with opportunities that may reduce the risk of injury [7]. Training load is one such modifiable risk factor that has garnered considerable research interest of late. Training load is typically the product of training intensity and duration and research suggests that an increase in training load is associated with greater incidences and severity of injury [11, 12]. However, an alternative view is that training loads that underprepare

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Key Points

The association between the acute:chronic workload ratio (ACWR) and non-contact injuries is supported by the literature as is its use as a valuable tool for monitoring training load as part of a larger scale multifaceted monitoring system that includes other proven methods.

The exponentially weighted moving average (EWMA) model is the more sensitive measure of ACWR in comparison to the rolling average (RA) model.

ACWR using both internal training load (ITL) and external training load (ETL) are associated with injury risk and practitioners can use either in isolation, or as a combination of both, as part of their monitoring practices.

The most commonly investigated ACWR time frame is 7:28 days but the most appropriate acute and chronic time periods may be dependent on the specific structure of the given sport.

The association between the ACWR and non-contact injury does not translate into the ability to predict the occurrence of an injury.

Practitioners need to be aware of the protective qualities of high chronic loads and physical strength and the detrimental effects of sudden spikes in acute training loads.

players for the demands of match play could place them at a greater risk of injury and hinder performance [13, 14]. The ideal training stimulus, therefore, is one that maximises performance by utilising an appropriate training load, whilst simultaneously limiting the negative consequences of under-training/overtraining [15].

Training load can be considered as either internal or external. Internal training load (ITL) is defined as the physiological and/or psychological stress imposed on the athlete during training and/or competition [16]. It is essentially the athlete's response to the external stimulus [17]. Common measures of ITL include subjective measures (self-reported) of rating of perceived exertion (RPE) and session rating of perceived exertion (sRPE) and objective measures of heart rate, blood lactate, oxygen consumption and creatine kinase. External training load (ETL) is the objectively measured work performed by the athlete during training and/or competition, independent of internal workloads [16]. Some commonly used assessments of ETL include speed, acceleration and distance covered. The importance and effectiveness of subjective internal measures of training load is well supported [18, 19]. Despite the merits of both ITL and ETL, it has been suggested that a combination of both methods may be most successful in monitoring training load [16, 17].

Training load can be further defined as absolute or relative. Absolute training loads are those accumulated in training and competition regardless of rate of load application, history of loading or fitness level. Relative training loads are those that do account for the rate of load application, history of loading and/or fitness level [20]. There is support for the absolute method of quantifying training load in relation to injury risk in several team sports [8]. However, the recent International Olympic Committee (IOC) consensus statement by Soligard et al. [20] states that relative load most closely associates with injury.

An important recent development in monitoring training load is the acute:chronic workload ratio (ACWR), a relative measure originating from the original work of Banister et al. [21] wherein it was known as the 'training stress balance', but more recently adapted to be calculated as a ratio by Hulin and colleagues [22] as opposed to a subtraction. The ACWR takes into account the current workload (acute) and the workload that an athlete has been prepared for (chronic) [22]. In essence, it is a model that may provide an index of athlete preparedness for training and/or competition [23]. The ACWR can be calculated using: (1) the rolling average model (RA) and (2) the exponentially weighted moving average model (EWMA). The RA model is calculated by dividing the acute (i.e.: rolling 7-day) workload by the chronic workload (i.e. average 28-day). The EWMA model, as explained by Williams et al. [24], assigns a decreasing weighting for each older load value in order to give greater weighting to the recent load performed by the athlete. There is much debate as to the best model of calculating ACWR with some suggestions that the RA model fails to account for the decaying nature of fitness and fatigue [24, 25]. Instead, the more sensitive EWMA model may be more suitable for modelling training loads [26]. Another consideration when calculating ACWR is whether acute and chronic workloads are mathematically coupled or uncoupled. In the coupled method, the acute load (the numerator) also comprises a part of the chronic load (the denominator) [27]. The alternative to this method is to exclude acute load periods in the calculation of chronic load (uncoupled). This inconsistency surrounding the calculation of ACWR requires further investigation and discussion.

1.1 Objectives

The primary objectives of this study are to (1) systematically review and evaluate the available evidence that examines the association between injury and the ACWR in team sport athletes and (2) investigate if sufficient evidence exists to determine the best method of application of the ACWR. In this present review, all reference to training load considers the load performed by an athlete in both training and competition. Recent reviews have examined the association between

training load and injury [7, 8, 28] but no such review has focused specifically on the ACWR. The use of the ACWR as a method of monitoring training load in order to mitigate the risk of injury has received growing interest in recent years. A deeper understanding of ACWR and training load management is necessary in an effort to potentially enhance the health and well-being of athletes through the reduction of overall injury risk. This would be best achieved through the systematic review of several studies that use comparable methodologies.

2 Methods

2.1 Literature Search

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were adhered to in this review [29]. The research question and inclusion/exclusion criteria (see Table 1) were established prior to conducting the literature search and were consistent with the objectives of the review. Studies were identified through a comprehensive search of the following six databases, from inception of the database until 16th November 2018: EMBASE, Medline, SPORTDiscus, SCOPUS, AMED and CINAHL. The databases were searched using combinations of the following keywords: ('injur* OR pain*) AND ("acute:chronic workload ratio" OR ACWR OR "training stress balance") OR ("exponentially weighted moving average" OR EWMA). Additionally, a search of the bibliographies of included studies and previous reviews was conducted to identify other potentially eligible articles, ensuring a comprehensive review. Studies that were in-press or not yet published at the time of the search were not included in this review. The titles and articles from the literature search were screened for suitability and all duplicates and articles clearly outside the scope of this review were removed. The full-text articles were then retrieved and checked against the inclusion/exclusion criteria by two independent reviewers (AG and ML).

2.2 Data Extraction

Extracted data from each article included author(s), year of publication, study design, sport, population sample size, level of competition, sex, age, study duration, injury definition, measurements/monitoring methods used for injury and training load (e.g. recorded by medical staff, sRPE, distance covered), load monitoring (internal and/or external), method of calculating ACWR (e.g. RA and/or EWMA, duration of acute and chronic periods), main statistical methods (e.g. odds ratios (OR), correlations, relative risk (RR) and main findings relating to the association between injuries and ACWR. Results were considered for meta-analysis. However due to the heterogeneity in the data extracted, the pooling of data was deemed inappropriate. For instance, the definition of injury, type of training load measured, methods of calculating ACWR and statistical analysis performed varied greatly between articles.

2.3 Quality Assessment

The methodological quality of the included studies was assessed by two reviewers (AG and ML) according to the Newcastle–Ottawa Scale (NOS) for Cohort Studies with some customisation carried out specific to the needs of this review [31]. The NOS is designed to assess for selection bias, comparability of the exposed and unexposed cohorts, outcome assessment and attrition bias and a quality score is calculated based on these three major components. A score between the ranges of 0–4, 0–2 and 0–3 was given to each

Table 1 Inclusion and exclusion criteria used in the systematic review

Inclusion	Exclusion
Team sport athletes participating in rugby union, rugby league, soccer, AFL, GAA, cricket, basketball, hockey, baseball or any other team sport discipline	Incomplete studies: abstracts, conference proceedings, editorials
Outcome measures to include *both* ACWR (or variation) and data regarding injury	Not peer-reviewed
Athletes are amateur or professional level	Not reported in English
Athletes are a minimum age of 18	Not original research
Prospective/retrospective cohort studies only[a]	Studies that measure outcomes not relevant to this systematic review
Injury surveillance of ≥ 3 months	
External or internal load	
Objective or subjective measures	

[a]As defined by the Cochrane Collaboration [30]

respective major component for a possible maximum score of 9. A higher score represents better methodological quality. The NOS was also selected for continuity purposes as it was used in two previous systematic reviews on related topics [7, 8]. For these reasons, the NOS was selected as the most appropriate quality assessment tool.

3 Results

3.1 Included Articles

A total of 210 potential articles were identified through the initial search of the databases, of which 102 were duplicates. After screening the titles, abstracts and full-texts, 19 articles met the inclusion criteria with an additional 3 articles included from the search of the bibliographies of included studies and previous reviews, resulting in a total of 22 articles. Figure 1 represents the different phases of the systematic search process.

3.2 Article Characteristics

3.2.1 Sporting Populations

All included articles were cohort studies addressing the association between the ACWR and injury. The sports included Australian Football ($n = 7$, 31.8%) [26, 32–37], soccer ($n = 6$, 27.3%) [23, 38–42], Rugby League ($n = 2$, 9.1%) [43, 44], Rugby Union ($n = 1$, 4.5%) [45], Gaelic Football ($n = 1$, 4.5%) [46], hurling ($n = 1$, 4.5%) [47], cricket ($n = 1$, 4.5%) [22], American Football ($n = 1$, 4.5%) [48], volleyball ($n = 1$, 4.5%) [49] and basketball ($n = 1$, 4.5%) [50]. Sample size ranged from 13 to 173 and all studies were conducted on male participants only. The mean age range was 22.0–26.7 years with two studies not reporting age [38, 45]. The level of the participants was reported as elite ($n = 16$, 72.7%) [22, 23, 26, 33–39, 41–44, 46, 49], professional ($n = 4$, 18.2%) [32, 40, 45, 50], amateur ($n = 1$, 4.5%) [47] and collegiate ($n = 1$, 4.5%) [48]. Ten studies (45.5%) [23, 36, 38, 42, 44–46, 48–50] monitored participants across 1 season, eight studies (36.4%) [26, 32, 34, 35, 40, 43, 44, 47] across 2 seasons, one study (4.5%) [39] across 3 seasons, two studies (9.1%) [33, 37] across 4 seasons and one study (4.5%) [22] across 5 seasons.

3.2.2 Injury Definition

Injury definition varied across the included articles with seven studies (31.8%) [23, 39–42, 46, 47] referring to the consensus statement of Fuller et al. [51] which defined injury as any physical complaint sustained by a player that resulted from a football match or football training, irrespective of the need for medical attention or time loss from football activities. Four studies (18.2%) [23, 41, 46, 47] referred to Brooks et al.'s [52] definition of any injury that prevents a player from taking a full part in all training and match play activities typically planned for that day for a period of greater than 24 h from midnight at the end of the day the injury was sustained. Four studies (18.2%) [32, 45–47] referred to Fuller et al.'s [53] definition of any physical complaint, which was caused by a transfer of energy that exceeded the body's ability to maintain its structural and/or functional integrity, that was sustained by a player during a rugby match or rugby training, irrespective of the need for medical attention or time-loss from rugby activities. Three studies (13.6%) [26, 35, 36] referred to Rogalski et al. [11] who defined injury as incidents resulting in a modified training programme, missed training sessions or games. Two studies (9.1%) [43, 44] referred to King et al.'s [54] injury definition of any injury that caused a player to seek medical or first aid treatment during or after a rugby league activity but did not lead to loss of further participation or non-selection for matches. Six studies (27.3%) [33, 34, 37, 38, 49, 50] referred to a different source and two studies (9.1%) [22, 48] used their own definition of injury.

3.2.3 Key Measures

Training load was measured using internal loads ($n = 9$, 40.9%), external loads ($n = 6$, 27.3%) or both ($n = 7$, 31.8%). All studies that measured ITL used the sum of sRPE and session duration with eight of these finding a statistically significant association between the ACWR and injury. All studies that utilised ETL reported a statistically significant association between the ACWR and injury. Most studies ($n = 18$, 81.8%) used the RA model to calculate ACWR, no study used the EWMA model exclusively and four studies (18.2%) used and compared both models. All 22 studies used the coupled method of calculating ACWR. Of the twenty-one studies using the RA model, 14 (66.7%) reported using an acute:chronic ratio of 7:28 days, one (4.8%) used a ratio of 3:21 days and six (28.6%) reported a range of various ratios. Of the five studies using the EWMA model, three (60%) used a ratio of 7:28 days, and two studies (40%) used a range of ratios. The summary of article characteristics is presented in Table 2.

3.3 Quality of Articles

The assessment of article quality had an overall median NOS score of 8 (range 5–9). The median score for 'selection' was 3 (range 2–4), the median score for 'comparability' was 2 (range 1–2) and the median score for 'outcome' was 3 (range 2–3). Only two studies accounted for the presence of injury at the start of the study and these studies also achieved a

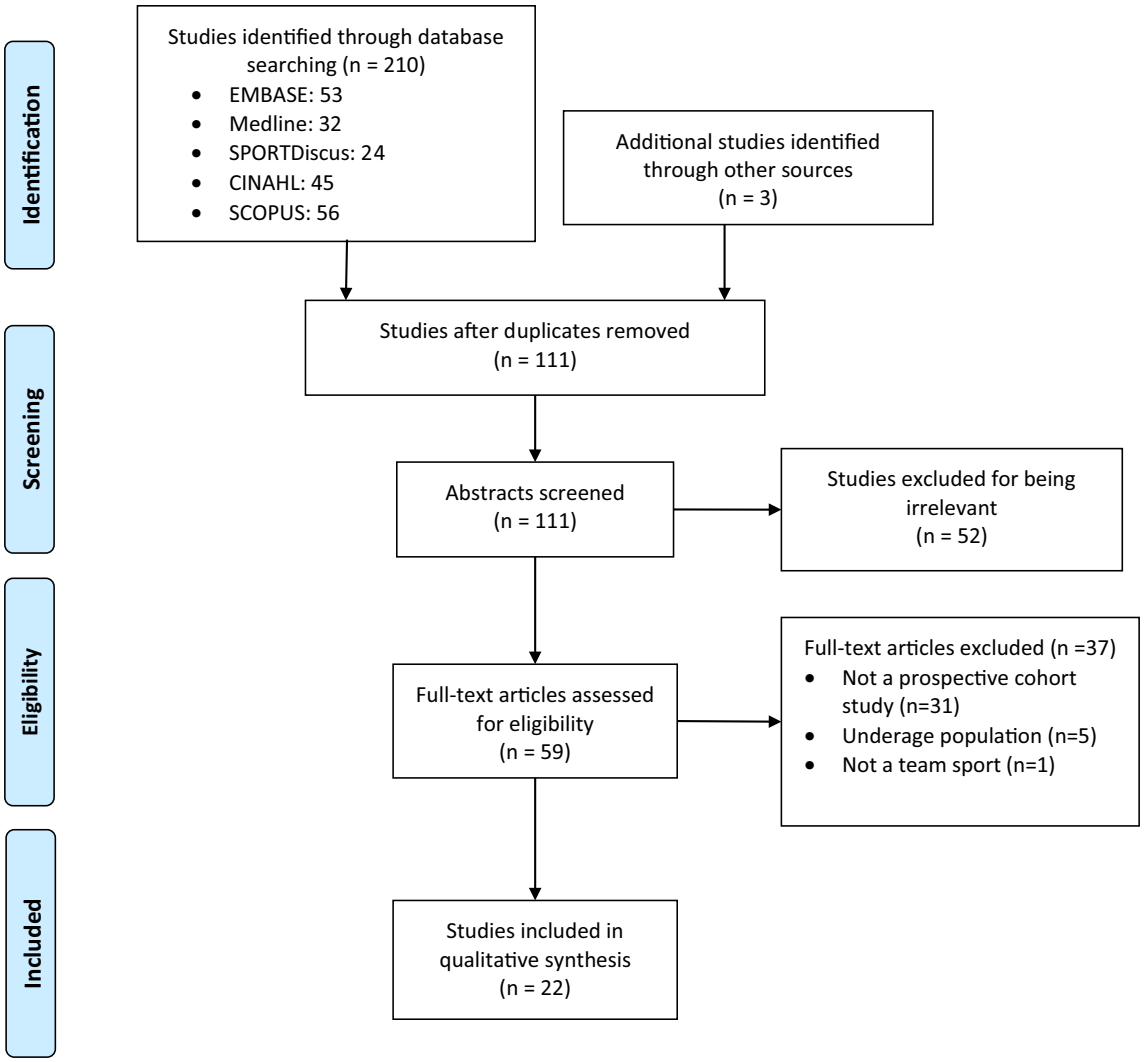

Fig. 1 PRISMA flowchart illustrating the selection process for inclusion of articles in the systematic review

maximum score of 9 [34, 38]. In accordance with the NOS, the quality of 21 articles was classified as 'good' (scale 7–9) and one study was classified as 'fair' (scale of 4–6) [31]. Table 3 represents the results of the NOS for the quality of the included articles. No included article reported a potential conflict of interest, 14 articles stated no conflict of interest while 8 articles did not report whether a potential conflict of interest was present.

4 Discussion

The primary aim of this systematic review was to examine the association between the occurrence of injury and the ACWR and to provide clarity and recommendations on methods of application in team sports. From the review of the included studies, it is apparent that there is evidence for this association. All but one study [45] found a significant association between the ACWR and injury. As nine articles (40.9%) from this review were published in 2018, it is evident that the ACWR is a method of training load monitoring that is growing in interest. Clarity is needed in relation to the reporting of injury and the most suitable method for the calculation of ACWR.

4.1 Reporting of Injuries

This review has identified conflicting definitions of what constitutes an injury as evident in Table 2. Multiple definitions to describe a constant may result in confusion and it has been suggested that it is meaningless to make comparisons between studies where injury definition is not the same [56]. The most commonly cited definition of injury is that of the consensus statement of Fuller et al. [51] which describes

Table 2 Summary of characteristics of articles investigating the association between injury and ACWR

Article	Population (*n*, level, sex, age)	Study duration	Injury definition	Load monitoring	ACWR model (ratio)	Main findings
Carey et al. [32] Prospective single-cohort	53, Australian Football, professional, male, 22.9 ± 4	2 seasons	Time-loss non-contact injuries [53]. Injuries recorded by medical staff	ITL and ETL. Load (total distance, high-speed distance, moderate speed distance, distance-load, player load) captured through GPS for field sessions and sRPE × duration for field sessions	RA across a range of ratios (acute 2–9 days, chronic 14, 18, 21, 24, 28, 32, 35 days)	The ratio of moderate speed running workload (18–24 km/h) at a 3:21 ratio best explained the injury likelihood in matches (R^2 = 0.79) and in the immediate 2 or 5 days following matches (R^2 = 0.7–0.82). The 3:21 ACWR discriminated between high-risk and low-risk athletes (relative risk (RR) =1.98–2.43). Using 6:21 ACRW window yielded similar results. The choice of acute time window significantly influenced model performance and appeared to reflect the competition and training schedule
Colby et al. [33] Prospective single-cohort	75, Australian Football, elite, male, 22.9 ± 3.4	4 seasons	Any lower body non-contact (intrinsic) injury resulting in matches missed [55]. Injuries recorded by club's senior physiotherapist	ITL and ETL. Load (total distance, high-speed distance) captured through GPS for field sessions and sRPE × duration for field sessions. Subjective wellness (fatigue, sleep quality, muscle soreness, stress levels, mood and perceived performance) collected via customised questionnaire. Weekly musculoskeletal screening	RA across a range of chronic windows (acute 7 days, chronic 14, 21, 28 days)	Very low cumulative chronic (2, 3, 4 week) workloads were associated with the greatest injury risk (univariate incidence rate ratio (IRR) = 1.71–2.16, 95% CI 1.10–4.52) in the subsequent week. In multivariate analysis, the interaction between a low chronic load and a very high distance (adjusted-IRR = 2.60, 95% CI 1.07–6.34) or low sRPE ACWR (adjusted-IRR = 2.52, 95% CI 1.01–6.29) was associated with increased injury risk. A clear association was evident between very low 2–4 week cumulative chronic loads (distance, On-Legs sRPE) and increased injury risk (IRR = 1.54–2.32, 95% CI 1.10–4.52), compared to moderate loads. A *U*-shaped association was evident with sprint (> 75% max speed) ACWR, indicating increased injury risk for both very low (IRR = 1.83, 95% CI 1.01–3.32) and very high (IRR = 1.90, 95% CI 1.01–3.58) ranges
Cross et al. [45] Prospective single-cohort	173, Rugby Union, professional, male, (age not reported)	1 season	Injuries that occurred in training or first- or second-team competitive matches that resulted in a time-loss of minimum 24 h [53]. Injuries were recorded by the medical personnel	ITL. Load captured through sRPE × duration for field and gym sessions	RA (7:28)	Generalised estimating equations showed a change in injury risk associated with a 2-SD increase in ACWR (172%) was unclear but may be a useful means of monitoring this aspect of loading

Table 2 (continued)

Article	Population (*n*, level, sex, age)	Study duration	Injury definition	Load monitoring	ACWR model (ratio)	Main findings
Delecroix et al. [38] Prospective cohort study (5 teams)	130, soccer, elite, male, (age not reported)	1 season	Any physical complaint sustained (upper or lower body) by a player that resulted from a non-contact incident in a soccer match or training, which made the player unable to participate in future soccer training or a match [56]. Injuries recorded by medical staff	ITL. Training load calculated using sRPE × duration for all sessions (including gym, bike etc.)	RA across a range of ratios (acute 7 days, chronic 7, 14, 21, 28 days)	Injury incidence was higher when the 7:28 ACWR was < 0.85 versus > 0.85 (RR = 1.31, 95% CI 1.02–1.70) and with a 7:21 ACWR > 1.30 versus < 1.30 (RR = 1.37, 95% CI 1.05–1.77). Importantly, none of the ACWR combinations showed high sensitivity or specificity
Esmaeili et al. [34] Prospective single-cohort	55, Australian Football, elite, male, 22.9 ± 3.9	2 seasons	Any training or match related incident that resulted in a missed match during the in-season phase or ≥6 days of modified training during the pre-season phase [57]. Injuries recorded by medical staff	ITL and ETL. Load (total distance, high-intensity running distance and PlayerLoad™) captured through GPS/accelerometer units for field sessions and sRPE × duration for all training sessions and matches	RA (7:28) and EWMA (7:28)	ACWRs were associated with increased risk of injuries. The effects for measures derived using EWMA were greater than those for RA. The effects of high ACWRs were overestimated by 10–15% before adjusting for previous injuries. Injury-prone players, identified via player identity in the mixed model, were at > 5 times higher risk of injuries compared to robust players (hazard ratio 5.4, 90% CI 3.6–12) despite adjusting for training load and previous injuries
Fanchini et al. [39] Prospective single-cohort	34, soccer, elite, male, 26 ± 5	3 seasons	Any non-contact injury that resulted in a player being unable to take a full part in future soccer training or match due to physical complaints [51]. Injuries recorded by medical staff	ITL. Training load calculated using sRPE × duration	RA across a range of ratios (acute 7 days, chronic 14, 21, 28 days)	Injury risk increased when 7:14 ACWR of 1.00–1.20, > 1.20 were compared to < 0.81 (odds ratio (OR), 90% CI 1.6, 0.79–3.29; 2.2, 1.03–4.74). Injury risk increased when comparing 7:21 ACWR of 1.01–1.23, > 1.23 vs. < 0.80 (OR, 90% CI 1.9, 0.9–3.8; 2.5, 1.2–5.4). Injury risk increased when comparing 7:21 ACWR of 0.78–1.02, 1.02–1.26, > 1.26 vs. < 0.78 (OR, 90% CI 2.4, 1.4–3.9; 3.3, 1.6–6.6; 3.5, 1.7–7.4)

Table 2 (continued)

Article	Population (*n*, level, sex, age)	Study duration	Injury definition	Load monitoring	ACWR model (ratio)	Main findings
Hulin et al. [22] Prospective cohort study (2 teams)	28, Cricket, elite, male, 26 ± 5	5 seasons	Any non-contact injury that resulted in a loss of either match time or greater than one training session over a 1-week period. Injuries recorded by medical staff	ITL and ETL. Training load calculated using balls bowled per week and sRPE × duration	RA (7:28)	ACWR was associated with an increased risk of injury in the week after exposure, for internal workload (relative risk (RR) = 2.2 (95% CI 1.91–2.53), $p = 0.009$), and external workload (RR = 2.1 (95% CI 1.81–2.44), $p = 0.01$). Fast bowlers with an internal workload ACWR of greater than 200% had a RR of injury of 4.5 (95% CI 3.43–5.90, $p = 0.009$) compared with those with an ACWR between 50% and 99%. Fast bowlers with an external workload ACWR of more than 200% had a RR of injury of 3.3 (95% CI 1.50–7.25, $p = 0.033$) in comparison to fast bowlers with an external workload ACWR between 50% and 99%
Hulin et al. [44] Prospective single-cohort	28, Rugby League, elite, male, 24.8 ± 3.4 years	2 seasons	Any non-contact time-loss injuries that resulted in a player being unable to complete full training, or missing match time [54]. Only injuries that were sustained in a match or reported subsequent to a match, prior to the next training session, were included. Injuries recorded by medical staff	ETL. Match and training load (absolute total distance run captured by GPS/micro-technology-units, between-match recovery days)	RA (7:28)	Compared with all other ratios, a very-high ACWR (≥ 2.11) demonstrated the greatest risk of injury in the current week (16.7% injury risk) and subsequent week (11.8% injury risk). High chronic workload (> 16 095 m) combined with a very high 2-week average ACWR (≥ 1.54) was associated with the greatest risk of injury (28.6% injury risk). High chronic workload combined with a moderate workload ratio (1.02–1.18) had the smallest risk of injury. Considering acute and chronic workloads in isolation (i.e. not as ratios) did not consistently predict injury risk

Table 2 (continued)

Article	Population (*n*, level, sex, age)	Study duration	Injury definition	Load monitoring	ACWR model (ratio)	Main findings
Hulin et al. [43] Prospective single-cohort	53, Rugby League, elite, male, 23.4 ± 3.5 years	2 seasons	Any time-loss injury that resulted in a player being unable to complete full training, or missing match time [44, 54, 58]. Only injuries that were sustained in a match or reported subsequent to a match, prior to the next training session, were included. Injuries recorded by medical staff	ETL. Load captured by GPS/micro-technology-units (distance (m) covered in field training and matches)	RA (7:28)	Players who had shorter recovery and ACWR ≥ 1.6, were 3.4–5.8 times likely to sustain a match injury than players with lower ACWRs (RR = 3.41–5.80 (90% CI 1.17–19.2); likelihood range 96–99%, very likely). ACWRs between 1.2 and 1.6 during short between-match recovery times demonstrated a greater risk of match injury than ratios between 1.0 and 1.2 (RR = 2.88 (90% CI 0.97–8.55); likelihood = 92%, likely)
Jaspers et al. [40] Prospective single-cohort	35, soccer, professional, male, 23.2 ± 3.7	2 seasons	Time-loss overuse injuries that resulted in a player being unable to take a full part in soccer training or match play [51]. Injuries recorded by medical staff	ITL and ETL. Load (total distance, high-speed distance, acceleration efforts and deceleration efforts) captured through GPS for field sessions and sRPE × duration for field and gym sessions	RA (7:28)	No most likely harmful effects were found. A most likely beneficial effect was found for a medium ACWR for RPE × duration (0.85–1.12). A very likely beneficial effect was found for a medium ACWR for number of decelerations (0.86–1.12). A high ACWR for distance covered at high speed should be avoided. Protective effects were found and a medium ACWR is recommended for number of accelerations, number of decelerations, and RPE × duration
Malone et al. [23] Prospective cohort study (2 teams)	48, soccer, elite, male, 25.3 ± 3.1	1 season	All injuries that prevented a player from taking full part in all training and match play activities typically planned for that day and prevented participation for a period greater than 24 h [51, 52, 59, 60]. Injuries recorded using a bespoke database	ITL. Load captured through sRPE × duration for training sessions (including gym and rehabilitation) and games	RA (7:28)	Binary logistic regression examining relative workload found players who exerted in-season ACWRs of > 1.00 to < 1.25 were at significantly lower risk of injury compared with those ≤0.85 (OR 0.28, 95% CI 0.08–1.26, $p = 0.001$)

Table 2 (continued)

Article	Population (*n*, level, sex, age)	Study duration	Injury definition	Load monitoring	ACWR model (ratio)	Main findings
Malone et al. [46] Prospective single-cohort	37, Gaelic Football, elite, male, 24.2 ± 2.9	1 season	All injuries that prevented a player from taking full part in all training and match play activities typically planned for that day and prevented participation for a period greater than 24 h [51–53]. Injuries recorded using a bespoke database	ITL. Load captured through sRPE × duration for training sessions (including gym) and games	RA (7:28)	Players who exerted in-season ACWRs of > 1.35 to 1.50 (OR = 0.88, 95% CI 0.28–4.66, $p = 0.006$) were at significantly lower risk of injury compared with the reference group of < 1.00. Players who exerted in-season ACWRs of > 2.00 were at increased injury risk (OR = 3.33, 95% CI 1.69–6.65, $p = 0.001$) compared with the reference group of < 1.00
Malone et al. [41] Prospective single-cohort	37, soccer, elite, male, 25 ± 3	1 season	All injuries that prevented a player from taking full part in all training and match play activities typically planned for that day and prevented participation for a period greater than 24 h [51, 52]. Injuries recorded using a bespoke database	ITL and ETL. Load (high speed running distance and sprint running distance) captured through GPS for field sessions and sRPE × duration for field and gym sessions	RA (3:21)	Players who had a high-speed running (> 14.4 km·h^{-1}) 3:21 ACWR of >1.25 and a sprint running distance (> 19.8 km·h^{-1}) 3:21 ACWR of > 1.35 were at increased risk of subsequent injury

Table 2 (continued)

Article	Population (*n*, level, sex, age)	Study duration	Injury definition	Load monitoring	ACWR model (ratio)	Main findings
Malone et al. [47] Prospective single-cohort	40, hurling, amateur, male, 26.2 ± 4.4	2 seasons	All injuries that prevented a player from taking full part in all training and match play activities typically planned for that day and prevented participation for a period greater than 24 h [51–53]. Injuries recorded using a bespoke database	ITL. Load captured through sRPE × duration for field and gym sessions	RA (7:28)	An ACWR between 0.90 and 1.30 was shown to offer protective effects, with the ratio explaining 60% of the variance associated with likelihood of subsequent injury. Slower 5 m (OR = 3.98, 95% CI 2.34–4.55), 10 m (OR = 2.78, 95% CI 1.32–3.14) and 20 m (OR = 4.55, 95% CI 2.12–4.98) athletes had increased injury risk when the weekly ACWR was ≥ 1.25
McCall et al. [42] Prospective cohort study (5 teams)	171, soccer, elite, male, 25.1 ± 4.9	1 season	An injury when a player was unable to fully participate in future soccer training or match due to physical complaints [51, 59]. Injuries recorded by medical staff	ITL. Training load calculated using sRPE × duration for all sessions (including gym, bike, recovery etc.) and matches	RA across a range of ratios (acute 7 days, chronic 7, 14, 21, 28 days)	ACWRs at 1:3 and 1:4 weeks were associated with non-contact injury ($P < 0.05$). Specifically, a greater risk of injury was found for players with a 7:28 ACWR of 0.97 to 1.38 (RR = 1.68; 95% CI 1.02–2.78, likely harmful) and > 1.38 (RR = 2.13; 95% CI 1.21–3.77, very likely harmful) compared to ACWR of 0.60 to 0.97. A 7:21 ACWR of > 1.42 compared with 0.59 to 0.97 displayed a 1.94 times higher risk of injury (RR = 1.90; 95% CI 1.08–3.36, very likely harmful). ACWR at both 7:28 and 7:21 showed poor predictive power (area under the curve (AUC) 0.53 to 0.58)
Murray et al. [35] Prospective single-cohort	59, Australian Football, elite, male, 23 ± 4	2 seasons	Any non-contact time-loss injury obtained during training or competition that resulted in a missed training session or missed game [11]. Injuries recorded by medical staff	ETL. Load (total distance, low-speed distance, moderate-speed distance, high-speed distance, very high-speed distance, and player load) captured through GPS for field sessions	RA (7:28)	RA ACWR was significantly associated with injury risk, with ACWR values > 2.0 associated with significantly greater risk of injury compared with those with values of 1.0–1.49 across a series of loading variables in both models. An ACWR of > 2.0 for total distance during the in-season was associated with a 5 to 8-fold greater injury risk in the current [(RR = 8.65, $p = 0.001$] and subsequent week (RR = 5.49, $p = 0.016$). Players with a high-speed distance ACWR of > 2.0 were 5–11 times more likely to sustain an injury in the current (RR = 11.62, $p = 0.006$) and subsequent week (RR = 5.10, $p = 0.014$)

Table 2 (continued)

Article	Population (*n*, level, sex, age)	Study duration	Injury definition	Load monitoring	ACWR model (ratio)	Main findings
Murray et al. [26] Prospective single-cohort	59, Australian Football, elite, male, 23.5 ± 4.4	2 seasons	As any non-contact 'time-loss' injury sustained during training or competition that resulted in a subsequent missed training session or game [11, 35]. Injuries recorded by medical staff	ETL. Load (total distance, low-speed distance, moderate-speed distance, high-speed distance, very high-speed distance, and player load) captured through GPS for field sessions	RA (7:28) and EWMA (7:28)	There were significant differences in the ACWR values between the RA and EWMA models for moderate (ACWR 1.0–1.49; $p = 0.021$), high (ACWR 1.50–1.99; $p = 0.012$) and very high (ACWR > 2.0; $p = 0.001$) ACWR ranges. Although both models demonstrated significant ($p < 0.05$) associations between a very high ACWR (> 2.0) and an increase in injury risk for total distance (RR = 6.52–21.28) and high-speed distance (RR = 5.87–13.43), the EWMA model was more sensitive for detecting this increased risk. The variance (R^2) in injury explained by each ACWR model was significantly ($p < 0.05$) greater using the EWMA model
Murray et al. [36] Single-cohort observational study	45, Australian Football, elite, male, 22 ± 3	1 season	Any non-contact 'time-loss' injury sustained during training or competition that resulted in a subsequent missed training session or game [11, 35]. Injuries recorded by medical staff	ETL. Load (total distance, high-speed distance, acceleration efforts and deceleration efforts) captured through GPS for field sessions	RA (7:28) and EWMA (7:28)	Large spikes in workload (i.e.: ACWR > 2.0), irrespective of model used resulted in an increased risk of injury at higher ACWR ranges. Faster players with a very-high-speed ACWR of > 2.0 had a greater risk of injury than those between 0.49 and 0.99 for both absolute (RR = 10.31; $p = 0.09$) and relative (RR = 4.28; $p = 0.13$) workloads
Sampson et al. [48] Retrospective single-cohort study	52, American Football, collegiate, male, 20.7 ± 1.5	1 season	All non-contact soft-tissue injuries. Injuries recorded by medical staff	ETL. Load (Playerload™) captured through GPS/IMU for field training sessions and matches	RA and EWMA across a range of ratios (7:14, 7:21 and 7:28)	7:21 EWMA ACWRs with a 3-day injury lag were most closely associated with injury ($R^2 = 0.54$). No specific ACWR experience or position interactions were identified
Stares et al. [37] Prospective single-cohort	70, Australian Football, elite, male, 22.9 ± 3.4	4 seasons	Intrinsic (non-contact) injury resulting in one or more matches being missed [61]. Injuries recorded by medical staff. Injuries recorded by medical staff	ITL and ETL. Load (total distance, high-speed distance) captured through GPS for field sessions and sRPE × duration for field sessions	RA across a range of ratios (acute 1–2 weeks, chronic 3–8 weeks)	ACWR from both ETL and ITL was significantly associated with injury risk ($p < 0.05$). Altering acute and/or chronic time frames did not improve the ability to detect high injury risk conditions above the commonly used 7:28 ACWR

Table 2 (continued)

Article	Population (*n*, level, sex, age)	Study duration	Injury definition	Load monitoring	ACWR model (ratio)	Main findings
Timoteo et al. [49] Prospective single-cohort	14, volleyball, elite, male, 26.7 ± 5.5	1 season	Any physical complaint that resulted in an inability of the player to fully take part in training and games, or if the player received assistance from the medical department team, even when participating in training or games [62]. Injuries recorded by medical staff	ITL. Training load calculated using sRPE × duration	RA (7:28)	There was a higher incidence of injuries ($p = 0.003$) and ACWR ($p < 0.001$) in the pre-season compared with the competitive period. Healthy players had lower ACWR ($p = 0.002$) compared with the injured players. The greater odds of injury was related to higher ACWR ($p = 0.014$)
Weiss et al. [50] Prospective single-cohort	13, basketball, professional, male, 24.4 ± 4.7	1 season	An injury was considered if an athlete reported any symptoms on any of the four questions of the Oslo Sports Trauma Research Centre Injury Questionnaire for each anatomical area [63]	ITL. Load captured through sRPE × duration for training sessions and games	RA (7:28)	Proportions of injured squad members at ACWRs between 1.0 and 1.49 were substantially less than those observed at all other ratios by clear small-to-moderate amounts. ACWRs ≤0.5, between 0.5 and 0.99, and ≥1.5 resulted in 1.5, 1.4 and 1.7 times more injured players, respectively. Comparisons between all other workload ratio ranges were trivial-to-small in magnitude and unclear

ITL internal training load, *ETL* external training load, *GPS* global positioning system, *RA* rolling averages, *ACWR* acute:chronic workload ratio, *sR*PE session rate of perceived exertion, *RR* risk ratio, *IRR* univariate incidence rate ratio, *CI* confidence interval, *SD* standard deviation, *EWMA* exponentially weighted moving averages, *OR* odds ratio, *RPE* rate of perceived exertion, *AUC* area under the curve, *IMU* inertial measurement units

Table 3 Quality assessment based on the Newcastle–Ottawa Scale (NOS) of articles in the systematic review

Article	Selection (/4)	Comparability (/2)	Outcome (/3)	Total (/9)
Carey et al. [32]	3	2	3	8
Colby et al. [33]	3	2	3	8
Cross et al. [45]	3	1	3	7
Delecroix et al. [38]	4	2	3	9
Esmaeili et al. [34]	4	2	3	9
Fanchini et al. [39]	3	2	3	8
Hulin et al. [22]	3	2	3	8
Hulin et al. [44]	3	1	3	7
Hulin et al. [43]	3	1	3	7
Jaspers et al. [40]	3	2	3	8
Malone et al. [23]	3	1	3	7
Malone et al. [46]	3	1	3	7
Malone et al. [41]	3	2	3	8
Malone et al. [47]	3	1	3	7
McCall et al. [42]	3	2	3	8
Murray et al. [35]	3	1	3	7
Murray et al. [26]	3	2	3	8
Murray et al. [36]	3	2	3	8
Sampson et al. [48]	3	2	3	8
Stares et al. [37]	3	2	3	8
Timoteo et al. [49]	3	1	3	7
Weiss et al. [50]	2	1	2	5
Median (range)	3 (2–4)	2 (1–2)	3 (2–3)	8 (5–9)

an injury as any physical complaint sustained by a player that results from a match or training, irrespective of the need for medical attention or time loss from sporting activities. The injury definition of Brooks et al. [52], referred to by four studies included in this review, is any injury that prevents a player from taking a full part in all training and match play activities typically planned for that day for a period of greater than 24 h from midnight at the end of the day the injury was sustained. These differences make comparisons between studies difficult.

Fuller et al. [53] further characterise a 'medical attention' injury as an injury that results in a player receiving medical attention, or a 'time loss' injury as an injury that results in a player being unable to take a full part in future training or match play. Despite the clear majority of the studies included in this review using medical staff to record injuries, the 'time-loss' definition is predominantly used. This is a strength of this review as the 'medical attention' definition increases the likelihood of the reporting of higher numbers of injuries which has the potential to include less reliable data [55, 64]. However, what constitutes a time-loss injury varies within the studies. For instance, the ≥ 6 days of modified training during the pre-season phase time-loss definition used by Esmaeili et al. [34] differs greatly from the 24 h used by several studies [23, 41, 45–47]. Hulin [65] highlights the need for clarity in the definition of injury, stating it has a definite influence on the results of ACWR examinations. To ensure studies that examine the association between injury and ACWR are comparable, ideally an international consensus definition of injury, such as that of Fuller et al. [53], should be used universally to ensure standardised reporting of results.

The majority of studies in this review recorded non-contact soft-tissue injuries only. It is argued by several authors and based on the work of Gabbett [66] that this type of injury is exclusively included as it is the only type of injury that is considered largely preventable and, therefore, more likely to be associated with the ACWR. However, contact injury should still be reported and analysed separately. Contact injuries have been found to represent a large proportion of injuries across many team sports [67–71] with some suggesting they are more severe than non-contact injuries [72]. Currently, there is no recognised or definitive definition of a contact injury. It could be argued that some contact injuries are unavoidable (e.g.: external force to the head caused by another player resulting in concussion) yet some may be more preventable (e.g.: external force resulting in soft tissue injury caused by fatigue reducing game awareness). Both these injuries will be classified as a contact injury by medical staff but their mechanisms are very different and this distinction should be clear. One of the fundamental principles of the ACWR is that higher chronic training loads protect

against injury providing the acute training load is similar to the chronic workload [22]. A robust athlete is one that is capable of enduring high training loads due to their highly developed physical qualities (e.g.: strength, speed) whilst an injury-prone athlete is more susceptible to injury due to the lack of these qualities [47]. It could be argued that the more robust an athlete, the greater the reduction in injury risk. Esmaeili et al. [34], for example, demonstrated that robust players were five times less likely to be injured compared to injury prone players. It is, therefore, conceivable that players with higher chronic training loads are less likely to become injured due to contact. This has been demonstrated by Bowen et al. [73] who found that contact injuries to elite youth soccer players are significantly related to an ACWR of 1.1–1.5 in various workload measures (total distance, number of decelerations and low-intensity distance). They concluded that by developing fitness levels and controlling fatigue, players may respond more rapidly and avoid the sudden and unpredictable movements preceding contact injury. Malone et al. [47] considered strength as a moderator of injury risk and found stronger athletes were more tolerable to larger week-to-week changes (> 550–1000 AU) in training load than weaker athletes (OR = 2.54–4.52). To take a practical example, an Australian Football player that is more physically robust due to increased strength levels brought on by greater training loads and training management is more likely to be protected against the large forces associated with the tackle in this contact sport. As none of the studies included in this review reported contact injuries independently, the association between the ACWR and contact injury is inconclusive and highlights a need for studies exploring this.

4.2 Calculating the Acute:Chronic Workload Ratio

The most commonly researched method of calculating ACWR in the studies in this review is the coupled RA model with most using an acute window of 7 days and a chronic window of 28 days. Although there is substantial support for this method within this literature, the EWMA model proposes an alternative that may yield more valid results.

4.2.1 Rolling Average Model

The RA model of calculating ACWR considers the association between training load and injury as linear, meaning each acute period within the aggregated chronic period is deemed equal. Despite this, there is an abundance of literature supporting its use [22, 33, 35, 37–39, 42–44, 49, 50]. In the early work, Hulin et al. [22] found that 7:28 RA ACWR in cricket players was associated with an increased injury risk in the following week, for internal workload (RR = 2.2, 95% CI 1.91–2.53), $p = 0.009$), and external workload (RR = 2.1, 95% CI 1.81–2.44), $p = 0.01$). Hulin et al. [44] found that acute and chronic training loads in isolation were not consistently associated with injury; however, when combined to form the ACWR there was a significant association. It was found that an ACWR of ≥ 2.11 demonstrated the greatest risk of injury in the current week (16.7% injury risk) and subsequent week (11.8% injury risk). Furthermore, the use of the RA model is supported in Australian Football [33, 35, 37], soccer [23, 38–42], Rugby League [43, 44], hurling [47], Gaelic Football [46], basketball [50] and volleyball [49] with the general consensus being that RA ACWR is an appropriate tool to monitor training load. Conversely, Cross et al. [45] found no association between RA ACWR and injury risk but still suggest it may be a useful means of monitoring training load and worthy of further investigation.

This increased risk of injury brought on by sudden increases in training load has also been seen using the ETL measure of sprint distance, defined as distance covered (m) above 75% of individual player maximum speed, in Australian Football players [33]. Colby et al. [33] found a *U*-shaped association indicating increased risk both very low (IRR = 1.83, 95% CI 1.01–3.32) and very high (IRR = 1.90, 95% CI 1.01–3.58) ranges. This *U*-shaped association is common throughout the studies with Malone et al. [23] suggesting ACWR using ITL between 1.00 and 1.25 is significantly associated with lower risk of injury compared to values greater than or less than this (OR 0.28, 95% CI 0.08–1.26, $p = 0.001$). Similarly, Jaspers et al. [40] advocate an ACWR between 0.85 and 1.12 for ITL and between 0.86 and 1.12 for ETL based on data collected from 35 professional male soccer players over two seasons.

It is evident that, across team sports, there is a protective value in having a high chronic training load and a moderate RA ACWR [44]. Importantly, Delecroix et al. [38] note that of the range of ACWRs (acute 7 days, chronic 7, 14, 21 and 28 days) investigated, none showed high sensitivity or specificity. This stresses the need for a more stringent model which may be seen in the form of the EWMA.

4.2.2 Exponentially Weighted Moving Average Model

Williams et al. [24] proposed the use of the EWMA model of calculating ACWR to address the concerns shared by Menaspà [25] that the RA method fails to account for the decaying nature of fitness and fatigue effects over time and the non-linearity of injury occurrence and workload. Since then, four studies included in this review have investigated this method with each providing strong support for its application [26, 32, 34, 36, 48]. Sampson et al. [48] examined both the RA and EWMA models with collegiate American Football players by plotting a range of ACWRs (7:14, 7:21 and 7:28 day) against non-contact injuries. They found that a 21-day EWMA ACWR presented a reasonable R^2 model fit

($R^2 = 0.54$) and was best able to explain non-contact injury risk. This is supported by the work of both Murray et al. [26] and Esmaeili et al. [34] who suggest that although both ACWR models are associated with increased injury risk, EWMA is a more sensitive measure and therefore more suitable than the traditional RA model ($p < 0.05$) which may underestimate the risk of injury at higher ACWRs. Four of the five studies that examined EWMA were conducted with elite Australian Football players; however, Sampson et al. [48] studied collegiate American Football players and found similar results reinforcing the validity of using this model.

From this review, it is evident that there is support for both models of calculating ACWR. Although more of the research has examined the RA model, it would appear that the EWMA is the more suitable measure, in part, due to its greater sensitivity [26]. What is important for practitioners to understand is that regardless of the model used, it is large and sudden increases in training load that are significantly associated with increased injury risk.

4.2.3 Internal and External Training Loads

Internal and external measures of training load have been thoroughly investigated with nine studies in this review having used ITL, six having used ETL and seven having used both. All but one study found significant findings in relation to the ACWR [45].The 13 studies that used ITL did so by means of the product of sRPE and time as described by Foster et al. [74], making it the most frequently reported method. The ETL measures amongst the studies included total distance, distance covered in various velocity zones, distance-load, acceleration efforts, deceleration efforts, player load and balls bowled per week. Hulin et al. [22] used balls bowled per week, during cricket, and the product of sRPE and duration when calculating ACWR and found both were significantly associated with an increased risk of injury in the week after exposure ($p = 0.01$). Jaspers et al. [40] also used both ETL (total distance, high-speed distance, acceleration efforts and deceleration efforts) and ITL (sRPE by duration). A medium ACWR (0.85–1.12) using the ITL measure showed a most likely beneficial effect and a medium ACWR (0.86–1.12) using the ETL measure showed a very likely beneficial effect. It is evident that ACWR using both ITL and ETL are significantly associated with injury risk and practitioners can use either in isolation, or as a combination of both.

4.2.4 Acute and Chronic Time Periods

Irrespective of the model used to calculate ACWR, the time period designated as the acute training period and the chronic training period has a significant effect on its association to injury [32]. The most commonly investigated ACWR is 7:28 days; however, seven studies examined the ACWR across a range of acute and chronic time frames; three in Australian Football [32, 33, 37], three in soccer [38, 39, 42] and one in American Football [48]. In soccer, the three studies found similar associations between 7:14, 7:21 and 7:28 day ACWR ratios. For instance, Fanchini et al. [39] found an OR of 2.98 for 7.14 ACWR, an OR of 2.46 for 7:21 ACWR and an OR of 2.91 for 7:28 ACWR, with each ratio being significantly associated with injury ($p \leq 0.01$). In general, soccer has regular fixtures one week apart giving justification to an acute window of 7 days [38, 39, 42]. The most suitable ratio may, therefore, depend on the specific structure of the given sport. This is supported by Carey et al. [32] who state that acute training load period has a strong influence over the ACWR's ability to inform injury risk and windows must be decided on a sport by sport basis. Further research is warranted into the optimal ACWR time frames across a multitude of sports. This may only be achieved by the analysis of identical ACWR metrics (e.g.: RA or EWMA, ITL or ETL, coupled or uncoupled), across an identical range of time frames, for the various sports.

4.2.5 The Mathematical Coupling Enigma

The twenty-two studies included in this review use a method of calculating the ACWR based on the coupled method. With this comes the possibility of spurious correlations meaning an association exists between two variables regardless of any true biological/physiological association between them [75]. In their editorial, Lolli et al. [27] generated data to simulate 4 weeks of ETL data for 1000 players reported in the study of Murray et al. [26]. The authors used the RA model with the most recent 7-day being the acute period and the 28-day period being the chronic load and found that if the acute training load was not included in the calculation of the chronic load then $r = 0.01$ (95% CI 0.05–0.07). They conclude that the numerator and denominator in the ACWR lead to bias inference as they are mathematically coupled causing a spurious correlation and a solution for this is to not include acute training load periods in the calculation of chronic training load. Conversely, in a study published subsequent to the literature search, Gabbett et al. [76] examined the association between coupled and uncoupled ACWR and injury risk in a cohort of 28 elite cricket fast bowlers. A nearly perfect relationship ($R^2 = 0.99$) was found between both methods demonstrating that higher ACWRs are associated with greater injury risk, regardless of whether acute and chronic workloads are coupled or uncoupled. This 'uncoupled' method is further discussed by Windt and Gabbett [77] where they define it as the ratio of the most recent week of work to the average of the three preceding weeks. In their comparison of both methods, the authors suggest that rather than promoting one approach over the other, researchers

must clearly detail it to facilitate interpretation and replication. It is clear, however, that there is a need for future research to examine the uncoupled method of calculating ACWR.

4.3 Association not Prediction

While the literature supports a significant association between the ACWR and non-contact injury, it is important to understand that this does not translate into the ability to predict the occurrence of an injury [78]. Shmueli [79] advocates a more careful use of terms such as prediction to reduce the effects of terminology on incorrect scientific conclusions. In order to determine if these associations with injury truly predict injury, different analysis should be used [80]. This is evident in the work of Fanchini et al. [39] who used a generalised estimating equations analysis to show a significant association between ACWR and injury risk in professional soccer players but also to show poor prediction ability using receiver operating characteristic curve (AUC ≤ 0.60). This research also highlights that, even at very high ACWRs, the number of false-positives greatly outweighs the true-positives at a rate of 274.7–15.3. Similarly, McCall et al. [42] found a significant association with non-contact injury occurrence but again these markers have poor predictive ability in isolation. Hulin and Gabbett [81] recently stated they regret using the words 'predict' and 'predictive' within their earlier studies [43, 44] but still advocate the use of the ACWR to monitor training load for individuals and teams.

4.4 Quality of Evidence

A very high level of quality was found across the 22 articles included in this review with just one receiving a score of less than 7 in the NOS. Only two studies [34, 38] reported information on whether the presence of injury existed at the start of the monitoring period highlighting the need for this to be conducted in future research. The 'comparability' component of the scale saw the most fluctuation in terms of scores due to variation in the variables reported in the studies (i.e.: ACWR model, internal/external training load, acute and chronic time frames). In order to increase methodological quality, future research should address these matters.

4.5 Limitations

Three of the 22 articles included in this review were not found by means of the database search but instead through the search of bibliographies. This would suggest a low risk of articles not being included. The search excluded non-English language articles, abstracts, posters, editorials and conference proceedings which could potentially result in inclusion bias.

The use of the NOS, as one of the Cochrane Collaboration recommended quality assessment tools, strengthens the methodology of this review. This is further reinforced by the use of a second author to validate the results of the scale as the inter-rater reliability of the NOS has been scrutinised [82]. The sensitivity of the NOS could, however, be questioned. For instance, the first question of the scale refers to the representativeness of the exposed cohort and both option (a) 'truly representative' and option (b) 'somewhat representative' receive the same score. This lack of sensitivity is a limitation of the scale.

A limitation highlighted across four of the studies included in this review is their methodology regarding missing data [34, 40, 42, 48]. In the event of missed training load, Sampson et al. [48] calculated the player's weekly training load average and added it to the dataset. Similarly, McCall et al. [42] used the player's season average for training or match loads. Currently, there is no best practice for dealing with missing data and McCall et al. [42] concluded that future research should explore methods of dealing with missing data.

The inconsistency in the reporting of injuries and the statistical analysis performed are additional limiting factors that restricted comparison between articles and subsequent meta-analyses.

4.6 Future Recommendations

The association between injury and the 7:28 RA ACWR across a range of sports has been well established. Future research needs to focus on the more sensitive EWMA model and across a range of time frames and across a larger range of sports. Additionally, research is warranted, similar to the work of Esmaeili et al. [34], that examines combinations of the EWMA model and other injury risk factors (i.e. injury and training history, subjective wellness, physical attributes, etc.). It has been suggested that the most suitable acute:chronic ratio is not necessarily sport specific but instead specific to the training and competitive nature of the sport, but evidence is needed to substantiate this [32].

The claims of Lolli et al. [27] that the 'uncoupled' calculation of the ACWR will avoid spurious correlation requires investigation and should be a focus of future research. Ultimately, researchers need to clearly detail the ACWR methodology (e.g.: RA or EWMA, coupled or uncoupled, etc.) and injury definition used to allow for both replication and clearer interpretation.

A clear disparity in the research is the absence of studies on female participants. Of the twenty-two studies in this review, ten sports were examined, at various levels, but all included male teams. Gender [sex] has been identified as a

risk factor associated with injury [83] so there is a need for female athletes to be examined in relation to the ACWR.

5 Conclusion

This systematic review is a comprehensive review of the literature examining the association between the ACWR and injury in team sport athletes. This is an emerging research area of interest with 22 articles published prior to 16th November 2018, 41% of which were published in 2018. The review highlights the disparity in the research in terms of the reporting of injuries and the application of ACWR and therefore warrants future research.

The findings of this review support the use of the ACWR as a valuable tool for monitoring training load as part of a larger scale multifaceted monitoring system that includes other proven methods. Although the association between ACWR and non-contact injury is evident from this review, it is important to understand that association does not translate to prediction and this term should be used with caution.

There is support for both the RA and EWMA models of calculating ACWR but it would appear that the EWMA is the more suitable measure, in part due to its greater sensitivity. Both ITL and ETL are suitable when calculating ACWR as both are significantly associated with injury risk and either can be used in isolation, or as a combination of both. The most suitable acute and chronic time periods to utilise in the calculation of the ACWR may be dependent on the specific structure of the given sport.

The application of this review for practitioners is the importance of understanding the intricacies of the ACWR before deciding on the best method of calculation for their desired population. It is also important to be aware of the protective qualities of high chronic loads and physical strength and the detrimental effects of sudden spikes in acute training loads.

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Compliance with Ethical Standards

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Conflict of interest Alan Griffin, Ian C. Kenny, Thomas M. Comyns and Mark Lyons declare that they have no potential conflicts of interest that are directly relevant to the content of this review.

Data Availability The datasets generated and/or analysed during the current study are available from the corresponding author on reasonable request.

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